Supplementary Material

Confidence Calibration and Uncertainty-Aware Prediction for Reliable Skin Lesion Classification

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This document extends the main paper with details that did not fit within its page limit. All numbers are generated from saved per-image predictions by the scripts in the code repository. E0/E1: ResNet-50 trained with plain or class-weighted cross-entropy; seed 42 is the model analysed in detail in the main paper. Source collections: HAM10000 [1], BCN_20000 [2] and MSK [3].

S1 Data, Sources and Duplicate Audit

Lesion identifiers are available for 23,247 of the 25,331 images. They cover all HAM10000 and BCN_20000 images and 819 MSK images; the remaining 2,084 MSK images have no identifier and were treated as single-image groups. The split uses 13,931 groups, 5,059 of which contain more than one image; no group occurs in two partitions.

Duplicate screen. Exact duplicates were found by MD5 hashes of the files and of the decoded 224×224 pixels: 50 groups of pixel-identical images, 4 of which cross partitions. Near-duplicates were found with a 64-bit DCT perceptual hash and the cosine similarity of ImageNet ResNet-50 embeddings. Candidate pairs had a hash distance ≤ 6 or a cosine ≥ 0.95, and a pair was flagged when the hash distance was ≤ 4 and the cosine ≥ 0.97. Within partitions, 97% of the 428 flagged pairs share a lesion identifier, which validates the rule. Across partitions it flags 10 pairs (7 near pixel-identical) involving 8 internal-test images. Visual inspection showed these to be re-uploads of the same image under different identifiers. Removing them gives balanced accuracy 0.565 (all images: 0.565). Removing every test image with any cross-partition candidate of cosine ≥ 0.97 (276 images) gives 0.576. Between the internal data and the official test set there are no exact duplicates and 2 flagged near-duplicates; excluding them leaves the official known-class balanced accuracy at 0.491 (all images: 0.491).

Dermoscopic artefacts. A dark circular vignette (the field stop of the dermatoscope) was detected automatically from the 224×224 images: an image is flagged when at least 3 of its 4 corner patches (16×16 px) have a mean grey level below 35/255. On a contact sheet of 24 randomly drawn internal-test images (12 flagged, 12 not flagged), the rule made no visible error; a larger manual validation was not done. Vignettes occur in 43% of BCN_20000, 3% of HAM10000 and 2% of MSK test images, and in 42% of official-test images. Table S1 suggests that the lower accuracy and error AUROC of vignetted images on the internal test split is mainly a source effect: within BCN_20000, vignetted images are classified as well as the others (0.655 vs. 0.638) and their errors are ranked no worse (AUROC 0.690 vs. 0.662). On the official test set, vignetted images are harder (accuracy 0.593 vs. 0.698) and contain more UNK lesions (33% vs. 19%), which suggests that its source mixture differs from the internal one. Rulers, ink markings and hair cannot be detected reliably without manual annotation and were not quantified.

Table S1: ResNet-50 E1 (seed 42) on images with and without a dark dermatoscope vignette (automatically detected: at least 3 of 4 corner patches with mean grey level below 35/255). Temperature and τ (85% validation coverage) from validation. UNK share: fraction of UNK images among all official-test images in that group.

Images	Vign.	n	Acc.	ECE$_T$	Cov.	Sel. acc.	Err. AUROC	UNK share
internal test, all sources	yes	543	0.656	0.059	0.847	0.707	0.697	–
	no	1980	0.742	0.053	0.894	0.785	0.779	–
internal test, BCN_20000 only	yes	510	0.655	0.067	0.847	0.704	0.690	–
	no	666	0.638	0.070	0.865	0.675	0.662	–
official test, known classes	yes	2328	0.593	0.029	0.796	0.655	0.722	0.331
	no	3863	0.698	0.031	0.854	0.753	0.776	0.188

S2 Training Configuration

Table S2: Training configuration (identical for all runs except the seed, and the input size in the replication of S8).

Input	224×224 RGB (320×320 in the replication of S8), bicubic resize, ImageNet mean/std
Augmentation (train)	horizontal flip, rotation $\pm 15^{\circ}$, scale $\pm 10\%$, translation $\pm 5\%$, brightness/contrast ± 0.1
Optimiser	AdamW, weight decay 10^{-4}, batch 32, gradient clipping 1.0, no label smoothing
Phase 1	backbone frozen, 10 epochs, learning rate 10^{-3}
Phase 2	last 30% of backbone parameter tensors unfrozen (ResNet-50: from the fifth bottleneck of layer3 onward, i.e. layer3.4–5 and all of layer4, 73% of the backbone weights; EfficientNet-B3: from within stage 6 onward, 78%), up to 15 epochs, learning rate 5×10^{-5}
Schedule / stopping	ReduceLROnPlateau on validation macro-F1 (factor 0.5, patience 3); early stopping on validation macro-F1 (patience 6)
Class weights (E1)	$w_c = N/(Kn_c)$ from the training split: NV 0.24, MEL 0.71, BCC 0.96, BKL 1.20, AK 3.70, SCC 5.01, VASC 12.2, DF 12.8
Hardware / software	NVIDIA RTX 4050 Laptop GPU (6 GB), PyTorch 2.11, CUDA 12.8; about 1 h per run (2 h at 320×320)

Seed 42 corresponds to the models analysed in detail in the main paper; seeds 43–46 were retrained with an identical protocol. For speed, training read images from a lossless cache of the 224×224 resize. This is bit-identical to resizing on the fly, as verified on 200 images.

S3 Checkpoint Rule and Seed-Level Statistics

The stored weights of the seed-42 models are those of the epoch at which early stopping triggered: epoch 24 for E1 (best validation macro-F1 at epoch 18) and epoch 21 for E0 (best at epoch 15). EfficientNet-B3 ran all 25 epochs, and its last epoch was also its best. For the new seeds both checkpoints were stored. The two rules give different weights whenever the best validation epoch is not the stopping epoch, which happened in 1 of 4 new E1 runs and 3 of 4 new E0 runs. Averaged over all new runs (identical weights included), the best-epoch weights had a lower temperature (E1: 2.308 vs. 2.530; E0: 2.491 vs. 3.278) and a lower uncalibrated ECE (E1: 0.114 vs. 0.131; E0: 0.145 vs. 0.190). Balanced accuracy (E1: 0.568 vs. 0.572; E0: 0.525 vs. 0.530), AURC (E1: 0.132 vs. 0.133; E0: 0.111 vs. 0.114) and calibrated NLL (E1: 0.816 vs. 0.815; E0: 0.755 vs. 0.765) were similar. Even the best-epoch weights needed a mean temperature above 2, so the over-confidence is not an artefact of the checkpoint rule. All main results use the stored-epoch rule so that all seeds are comparable.

Fig. S1 shows the validation curves of all 17 runs. In 17 of them the validation loss is lowest at epochs 10–15, shortly after the backbone is unfrozen, and rises steeply afterwards, while validation macro-F1 keeps improving. Checkpoints chosen on macro-F1 therefore have 1.15–2.76 times the minimum validation loss, which is why every model needs a temperature well above 1. We did not test checkpoint selection on validation loss; we kept macro-F1, the original checkpoint criterion, and rely on post-hoc calibration.

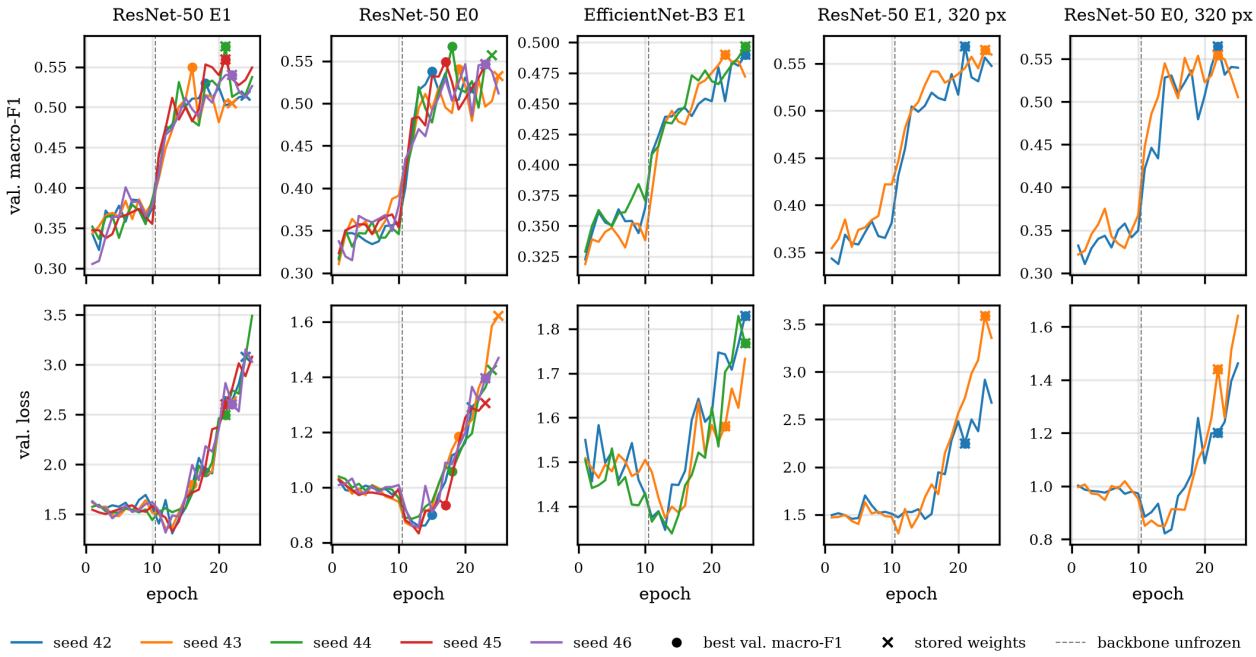


Figure S1: Validation macro-F1 (top) and validation loss (bottom) per epoch for every run. Dots: best validation macro-F1; crosses: epoch of the stored weights (they coincide when early stopping did not trigger). Dashed line: backbone unfrozen. Seed-42 curves at 224×224 are parsed from the original training logs. For E1 the validation loss is class-weighted.

Table S3: Mean $\pm$ SD over independently trained seeds (seed 42 = the run analysed in detail; further seeds trained with the identical protocol). All post-hoc maps and thresholds fitted on validation. Last column: mean per-seed difference, with (number of seeds where E1 $>$ E0 / seeds paired, paired t-test p). †Official ISIC 2019 test set, never used during development (8 known classes; UNK = out-of-distribution).

Metric (internal test unless marked)	E0 ResNet-50 (n=5)	E1 ResNet-50 (n=5)	E1 EffNet-B3 (n=3)	E1$-$E0 paired
Accuracy	0.726 ± 0.005	0.719 ± 0.010	0.684 ± 0.005	-0.008 (2/5, $p = 0.15$)
Balanced accuracy	0.532 ± 0.006	0.571 ± 0.013	0.543 ± 0.008	$+0.039$ (5/5, $p = 0.004$)
Macro-F1	0.543 ± 0.011	0.555 ± 0.012	0.502 ± 0.013	$+0.011$ (4/5, $p = 0.09$)
ROC-AUC	0.926 ± 0.003	0.922 ± 0.005	0.920 ± 0.001	-0.004 (1/5, $p = 0.08$)
ECE (uncalibrated)	0.188 ± 0.009	0.135 ± 0.019	0.083 ± 0.004	-0.053 (0/5, $p = 0.009$)
ECE (T)	0.026 ± 0.005	0.049 ± 0.011	0.043 ± 0.011	$+0.024$ (5/5, $p = 0.009$)
NLL (T)	0.762 ± 0.009	0.814 ± 0.005	0.870 ± 0.016	$+0.052$ (5/5, $p < 0.001$)
Brier (T)	0.380 ± 0.003	0.401 ± 0.006	0.438 ± 0.006	$+0.021$ (5/5, $p = 0.002$)
Temperature T	3.209 ± 0.247	2.606 ± 0.222	1.660 ± 0.100	-0.603 (0/5, $p = 0.02$)
Error AUROC, MSP	0.792 ± 0.006	0.762 ± 0.006	0.761 ± 0.012	-0.030 (0/5, $p = 0.001$)
Error AUROC, MC entropy	0.778 ± 0.008	0.750 ± 0.007	0.761 ± 0.011	-0.028 (0/5, $p < 0.001$)
AURC, MSP	0.113 ± 0.004	0.133 ± 0.005	0.152 ± 0.006	$+0.020$ (5/5, $p < 0.001$)
Coverage at val. 85%	0.870 ± 0.010	0.870 ± 0.013	0.869 ± 0.004	$+0.000$ (2/5, $p = 0.99$)
Selective accuracy	0.776 ± 0.004	0.765 ± 0.009	0.731 ± 0.004	-0.012 (1/5, $p = 0.07$)
Official test bal. acc.†	0.472 ± 0.012	0.508 ± 0.014	0.523 ± 0.002	$+0.037$ (5/5, $p = 0.02$)
Official test ECE (T)†	0.021 ± 0.004	0.033 ± 0.009	0.019 ± 0.006	$+0.012$ (4/5, $p = 0.05$)
UNK detection AUROC (MSP)†	0.664 ± 0.019	0.624 ± 0.013	0.618 ± 0.004	-0.041 (0/5, $p = 0.007$)

Table S4: Which model is best depends on the reliability dimension (means over seeds; internal test unless stated).

Dimension	R50-E0	R50-E1	B3-E1	Best
Balanced accuracy	0.532	0.571	0.543	R50-E1
Accuracy	0.726	0.719	0.684	R50-E0
Macro-F1	0.543	0.555	0.502	R50-E1
ROC-AUC	0.926	0.922	0.920	R50-E0
ECE, uncalibrated	0.188	0.135	0.083	B3-E1
ECE after T	0.026	0.049	0.043	R50-E0
NLL after T	0.762	0.814	0.870	R50-E0
Error AUROC (MSP)	0.792	0.762	0.761	R50-E0
AURC (MSP)	0.113	0.133	0.152	R50-E0
Official test bal. acc.	0.472	0.508	0.523	B3-E1
Official test ECE$_T$	0.021	0.033	0.019	B3-E1
UNK detection AUROC	0.664	0.624	0.618	R50-E0

S4 Calibration Details

ECE uses 15 equal-width bins; adaptive ECE (aECE) uses 15 equal-mass bins and class-wise ECE (cwECE) averages the ECE of each class probability [4, 5]. Prior correction subtracts $\log w_c$ from the logits of E1 (and logit adjustment adds it to E0) following [6]. Calibration-aware training losses such as focal loss [7] were not evaluated.

Table S5: Calibration on the internal test split (seed 42). All maps are fitted on validation only. ECE: 15 equal-width bins; aECE: 15 equal-mass bins; cwECE: class-wise ECE. Vector scaling and prior correction change the decision rule, hence the last column.

Model	Post-hoc map	ECE	aECE	cwECE	Brier	NLL	Bal. acc.
ResNet-50 E1	none	0.149	0.148	0.042	0.425	1.264	0.565
	temperature	0.050	0.047	0.032	0.394	0.811	0.565
	vector scaling	0.038	0.041	0.015	0.375	0.768	0.540
	prior correction	0.139	0.139	0.038	0.409	1.305	0.533
	prior corr. + T	0.065	0.066	0.031	0.384	0.803	0.533
ResNet-50 E0	none	0.179	0.177	0.046	0.438	1.229	0.537
	temperature	0.031	0.029	0.014	0.375	0.753	0.537
	vector scaling	0.025	0.023	0.011	0.375	0.751	0.505
EffNet-B3 E1	none	0.086	0.087	0.041	0.452	0.990	0.536
	temperature	0.055	0.057	0.038	0.443	0.882	0.536
	vector scaling	0.024	0.024	0.014	0.392	0.792	0.444
	prior correction	0.092	0.089	0.026	0.409	0.936	0.457
	prior corr. + T	0.046	0.048	0.026	0.401	0.819	0.457

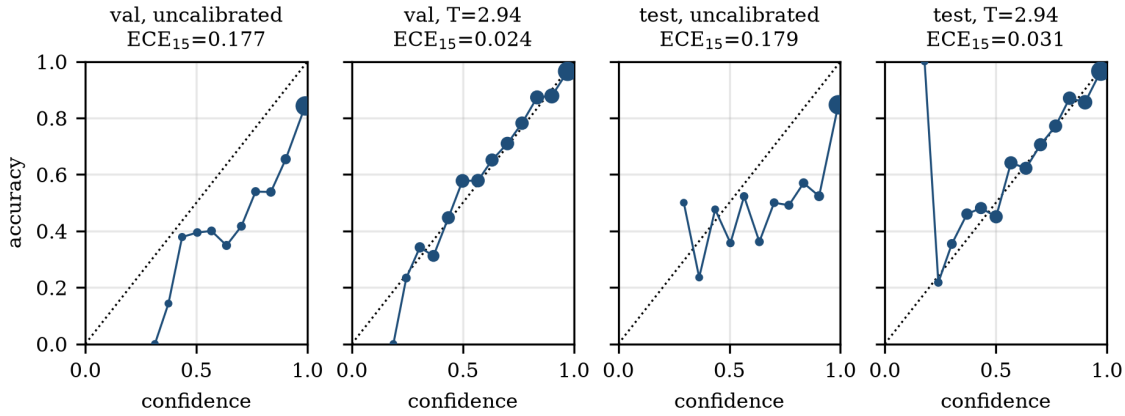


Figure S2: Reliability diagrams of ResNet-50 E0 (seed 42) on validation and internal test, before and after temperature scaling.

S5 Selective Prediction Details

Table S6: Validation-locked selective prediction (ResNet-50 E1, seed 42). Thresholds are chosen on validation for a target validation coverage and applied unchanged to test. Rej. acc.: accuracy among rejected test cases; MEL sens.: melanoma sensitivity among accepted cases (all cases: 0.693).

Score	Target	τ	Val. cov.	Test cov.	Sel. acc.	Rej. acc.	MEL sens.	MEL rej.
MSP (cal.)	0.70	0.524	0.690	0.723	0.812	0.493	0.756	147
MSP (cal.)	0.80	0.463	0.802	0.841	0.786	0.397	0.742	72
MSP (cal.)	0.85	0.429	0.851	0.884	0.769	0.380	0.733	53
MSP (cal.)	0.90	0.391	0.900	0.922	0.752	0.383	0.711	32
MSP (cal.)	0.95	0.348	0.941	0.950	0.744	0.333	0.706	18
MSP (raw)	0.70	0.798	0.691	0.734	0.809	0.487	0.790	161
MSP (raw)	0.80	0.674	0.801	0.836	0.781	0.434	0.755	95
MSP (raw)	0.85	0.616	0.851	0.878	0.764	0.432	0.732	66
MSP (raw)	0.90	0.564	0.901	0.912	0.756	0.395	0.728	47
MSP (raw)	0.95	0.518	0.941	0.947	0.745	0.338	0.709	18
MC entropy	0.70	1.187	0.700	0.746	0.798	0.506	0.741	113
MC entropy	0.80	1.330	0.800	0.828	0.773	0.488	0.726	72
MC entropy	0.85	1.419	0.850	0.874	0.763	0.455	0.719	53
MC entropy	0.90	1.517	0.900	0.922	0.749	0.423	0.704	32
MC entropy	0.95	1.622	0.950	0.956	0.737	0.432	0.697	16

Table S7: Risk-controlled thresholds (SGR, Geifman & El-Yaniv) fitted on validation with $\delta = 0.05$ and applied to test.

Model	r^*	θ	Bound	Val. cov.	Test cov.	Test risk
ResNet-50 E1	0.10	0.770	0.099	0.342	0.386	0.086
ResNet-50 E1	0.15	0.643	0.150	0.503	0.555	0.136
ResNet-50 E1	0.20	0.532	0.200	0.674	0.710	0.181
ResNet-50 E1	0.25	0.457	0.250	0.813	0.848	0.218
ResNet-50 E0	0.10	0.806	0.099	0.421	0.434	0.079
ResNet-50 E0	0.15	0.667	0.150	0.588	0.607	0.130
ResNet-50 E0	0.20	0.548	0.200	0.731	0.753	0.172
ResNet-50 E0	0.25	0.417	0.250	0.886	0.897	0.229
EffNet-B3 E1	0.10	0.791	0.100	0.299	0.327	0.117
EffNet-B3 E1	0.15	0.651	0.150	0.457	0.487	0.151
EffNet-B3 E1	0.20	0.576	0.200	0.560	0.584	0.179
EffNet-B3 E1	0.25	0.496	0.250	0.689	0.711	0.212

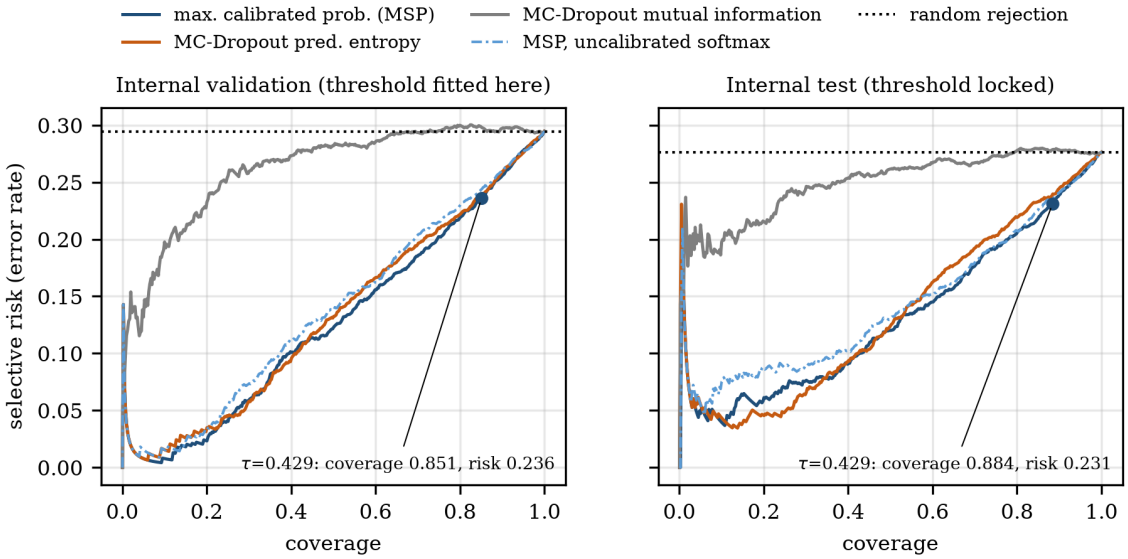


Figure S3: Risk–coverage curves of ResNet-50 E1 (seed 42) on validation, where the threshold is fitted, and on the internal test split, where it is locked, for calibrated MSP, uncalibrated (raw softmax) MSP, MC predictive entropy and MC mutual information, with random rejection as reference.

S6 Uncertainty Scores: Full Table

Mutual information and expected entropy follow the standard decomposition of predictive entropy [8]. Conformal sets (main paper) use the LAC score [9], the adaptive APS score [10] and a class-conditional (Mondrian) variant of LAC. Selective risk is summarised by AURC and E-AURC [11], and risk-controlled thresholds use SGR [12]. All calibrated scores use temperature-scaled logits; for MC Dropout each of the 30 passes is scaled by T before averaging. Uncalibrated variants (raw logits) are listed for MSP, softmax entropy and MC entropy. A single temperature keeps every predicted class but can reorder images by confidence, because the maximum probability depends on all logit gaps; the AUROCs of raw and calibrated scores therefore differ.

Table S8: Error detection on the internal test split (ResNet-50 E1, seed 42). Target: errors of the deployed (deterministic) classifier. Δ: paired lesion-cluster bootstrap difference to calibrated MSP. pAUROC: standardised partial AUROC for FPR ≤ 0.2. AURC: area under the risk–coverage curve (random rejection = 0.276).

Score	AUROC	Δ vs MSP [95% CI]	pAUROC	AURC
MSP $(1 - \max_k p_k)$, calibrated	0.761	–	0.627	0.134
MSP, uncalibrated	0.745	−0.016 [−0.025, −0.007]	0.621	0.144
Softmax entropy, calibrated	0.750	−0.010 [−0.022, +0.001]	0.601	0.134
MC predictive entropy (paper)	0.745	−0.016 [−0.029, −0.003]	0.598	0.136
MC expected entropy (aleatoric)	0.747	−0.014 [−0.027, −0.002]	0.600	0.136
MC mutual information (epistemic)	0.528	−0.232 [−0.276, −0.189]	0.497	0.249
MC predictive variance	0.620	−0.141 [−0.178, −0.107]	0.534	0.197
MC mean-probability MSP	0.761	+0.000 [−0.003, +0.003]	0.625	0.134
Softmax entropy, uncalibrated	0.750	−0.011 [−0.017, −0.005]	0.630	0.142
MC predictive entropy, uncalibrated	0.756	−0.004 [−0.011, +0.002]	0.627	0.139
TTA predictive entropy	0.760	−0.000 [−0.016, +0.016]	0.615	0.130
TTA mutual information	0.705	−0.056 [−0.085, −0.028]	0.580	0.154

S7 Grad-CAM Checks

The main paper shows four curated Grad-CAM examples [13]. We applied two quantitative checks to 300 random test images, at the original layer (`conv3` of the last bottleneck) and at the output of `layer4`.

- *Cascading parameter randomisation* [14]. Re-initialising only the head already drops the rank correlation between original and randomised maps to 0.04, so the maps depend on the trained weights (Table S9).
- *Deletion/insertion* [15]. Grad-CAM clearly beats a random map. For deletion, however, it is no better than a fixed centre-Gaussian prior that ignores the image (AUC 0.538 vs. 0.538, Wilcoxon $p = 0.861$); for insertion it is only modestly better (0.754 vs. 0.723, $p = 0.002$) (Table S10).

On average, 55.5% of the map mass falls in the central quarter of the image (Fig. S4). Maps of uncertain predictions are more concentrated (Spearman $\rho = -0.58$ between map entropy and MC entropy) and often lie off the lesion (Fig. S5). As an error detector, map concentration reaches an AUROC of only 0.627. Grad-CAM is therefore not used as reliability evidence in the main paper.

Table S9: Cascading model-parameter randomisation (300 random test images): similarity between Grad-CAM maps of the trained and the progressively randomised network. Values near 0 indicate model-sensitive maps.

Randomised	Spearman ρ	Top-10% IoU
up to head	0.043	0.166
up to layer4	0.115	0.078
up to layer3	−0.034	0.048
up to layer2	0.022	0.053
up to layer1	−0.094	0.041

Table S10: Deletion/insertion faithfulness (300 random test images; probability of the predicted class as salient pixels are blurred out / restored in 5% steps). Lower deletion and higher insertion AUC are better.

Saliency	Deletion AUC	Insertion AUC
Grad-CAM (paper layer)	0.538	0.754
Grad-CAM (layer4 output)	0.533	0.760
Dataset-mean Grad-CAM	0.540	0.731
Centre Gaussian prior	0.538	0.723
Random map	0.586	0.579

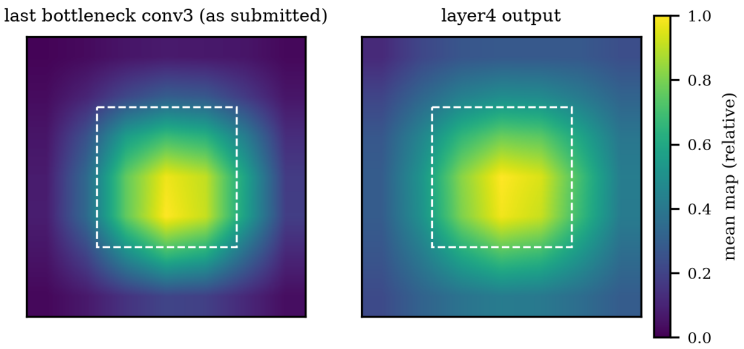


Figure S4: Mean Grad-CAM map of ResNet-50 E1 over the 300 random test images, at the layer used in the main paper and at the output of layer4, normalised to its maximum. Dashed square: central quarter of the image. On average the maps concentrate in the image centre, where lesions also usually lie, which is consistent with a fixed centre-Gaussian prior being about as faithful on deletion.

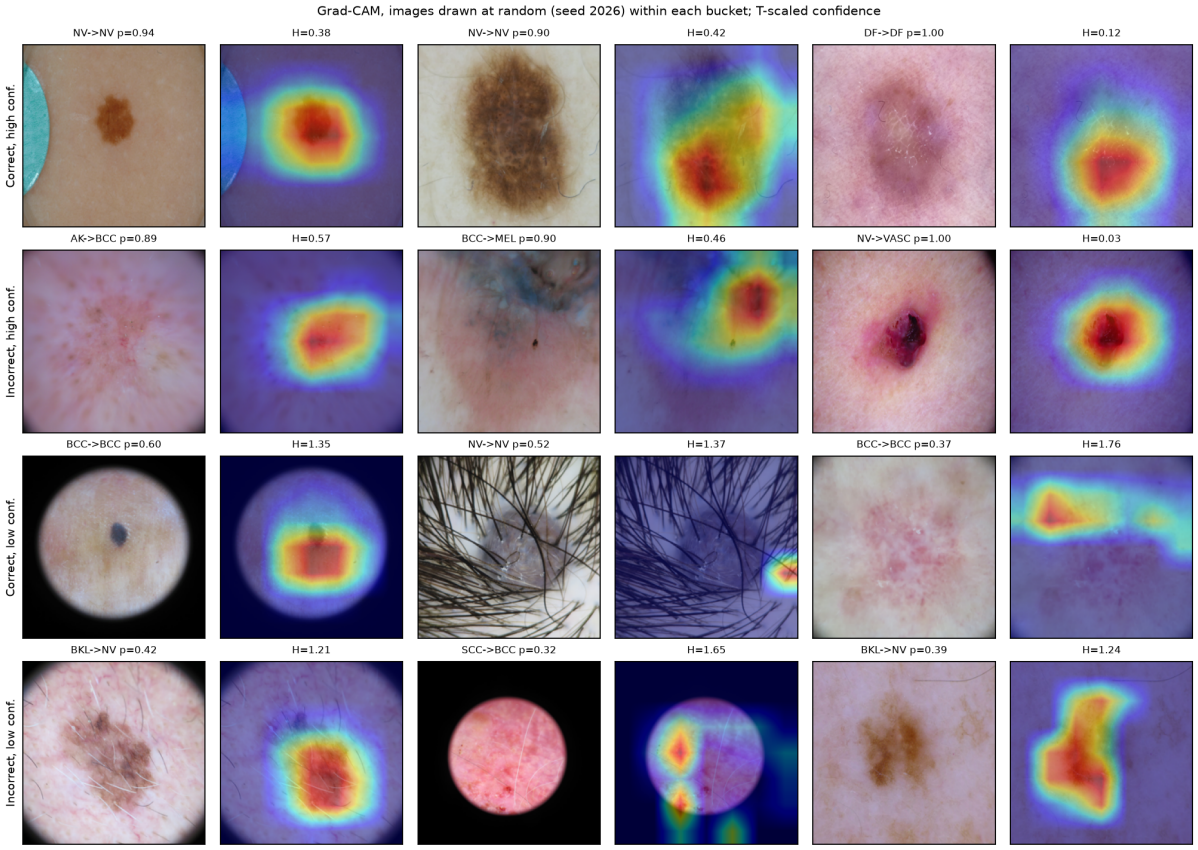


Figure S5: Grad-CAM maps for test images drawn at random (fixed seed) from each correctness/confidence bucket of ResNet-50 E1. Titles: true→predicted class, calibrated confidence and MC entropy H.

S8 Higher-Resolution Replication

To test whether the conclusions depend on the 224×224 input, ResNet-50 E0 and E1 were retrained at 320×320 (seeds 42 and 43) with otherwise identical settings (Table S11).

With 2 paired seeds this replication is small, so it is a consistency check rather than a separate result. Relative to the same seeds at 224×224, the higher resolution changed balanced accuracy by +0.039 (E0) and +0.016 (E1) and AURC by -0.006 and -0.014, while the calibrated ECE changed by +0.027 and +0.009. The reliability cost of class weighting recurred in 2 of 2 seeds: after temperature scaling, E1 had a higher NLL (+0.046), a larger AURC (+0.011), a lower error AUROC (-0.015) and a lower UNK detection AUROC on the official test set (-0.070) than E0. Its balanced-accuracy gain was not consistent (per seed +0.059 and -0.011). Logit adjustment of the 320×320 E0 model reached a balanced

Table S11: ResNet-50 at 224×224 vs. 320×320 px (mean±SD over seeds; internal test unless stated). Same protocol; only the input resolution changes.

Metric	E0 224 (n=5)	E1 224 (n=5)	E0 320 (n=2)	E1 320 (n=2)
Balanced accuracy	0.532±0.006	0.571±0.013	0.572±0.017	0.596±0.032
Accuracy	0.726±0.005	0.719±0.010	0.749±0.003	0.740±0.015
ECE, uncalibrated	0.188±0.009	0.135±0.019	0.152±0.001	0.129±0.020
ECE, after T	0.026±0.005	0.049±0.011	0.052±0.005	0.057±0.001
NLL, after T	0.762±0.009	0.814±0.005	0.727±0.009	0.774±0.027
Temperature	3.209±0.247	2.606±0.222	3.094±0.245	2.733±0.465
Error AUROC (MSP)	0.792±0.006	0.762±0.006	0.776±0.004	0.761±0.011
Error AUROC (MC entropy)	0.778±0.008	0.750±0.007	0.753±0.009	0.753±0.017
AURC (MSP)	0.113±0.004	0.133±0.005	0.107±0.004	0.119±0.011
Official bal. acc.	0.472±0.012	0.508±0.014	0.515±0.011	0.530±0.029
Official UNK AUROC (MSP)	0.664±0.019	0.624±0.013	0.696±0.028	0.626±0.012
Official UNK AUROC (MC MI)	0.699±0.017	0.666±0.010	0.699±0.004	0.680±0.004

accuracy of 0.616 (E1: 0.596), with a lower ECE (-0.013) and a higher error AUROC (+0.022) than E1 but a similar AURC (+0.003).

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